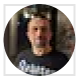
Time Series with Konrad: episode 4

Hierarchical time series



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Hierarchical structures show up everywhere: organizations, geographies, product catalogs - so, inevitably, also in time series data. Sales roll up from stores to cities to countries, electricity demand aggregates from households to substations to national grids, tourism flows sum from attractions to regions to entire countries… You get the idea.

When we forecast such data, we are not dealing with a collection of independent time series. What we’ve got are **multiple, interdependent views of the same future**. This is the core idea behind *hierarchical time series forecasting*.

In this episode, we will explore how to produce forecasts that are both *accurate* and *coherent*. We start with simple, intuitive strategies (Bottom-Up, Top-Down, and Middle-Out), understand their strengths and weaknesses, and then move to modern optimal reconciliation methods that use information **from the entire hierarchy at once**.

📓 *All code lives in the companion notebook:* https://github.com/tng-konrad/time-series-with-Konrad/blob/main/m04-hierarchical-forecasting.ipynb

Hierarchical models

The defining feature of hierarchical forecasting is the requirement for consistency **across all levels of the hierarchy**: this means that the sum of the forecasts at the lower levels must equal the forecasts at the higher levels. For instance, the sum of forecasted sales for all cities in a region must mathematically equal the forecasted sales for the region as a whole.

While this sounds like a trivial accounting identity, it poses a decidedly non-trivial statistical challenge: when forecasts are generated independently for each series in the hierarchy - a.k.a. “base forecasting” - they never sum up correctly due to model estimation errors, varying noise levels, and differing aggregation dynamics.

This is a problem: sticking with the sales example, if the sum of regional demand forecasts exceeds the national demand forecast, the logistics department and the procurement department will be working with conflicting numbers.

The primary objective of hierarchical forecasting is not only to predict future values but to reconcile these predictions into a mathematically consistent structure that maximizes accuracy across all levels simultaneously.

Hierarchical vs grouped

It is useful to distinguish between strict hierarchical structures and grouped structures, as the topology of the data influences the choice of reconciliation strategy (even though they are often treated under the same umbrella).

Hierarchical Time Series follow a nested, tree-like structure. A classical example is geography:

Total → Country → State → Region → Store

In this topology, every node has a single unique parent: a store in ‘California’ cannot also belong to ‘Texas’. The aggregation logic is unidirectional and unambiguous.

Grouped Time Series involve crossed aggregation structures: consider a bicycle manufacturer that categorizes sales by *Geography* (State/Region) and by *Product Type* (Road/Mountain/Hybrid).

- One hierarchy aggregates: Total → State → Region.
- Another aggregates: Total → Product Type → Sub-type.
- The data can also be sliced by the intersection: Total → State → Product Type.
- Here, the structure is a lattice, not a a tree. A “Mountain Bike” sale belongs to the “Mountain” product group *and* the “California” geographic group simultaneously.

In the remainder of this post, we will focus on the hierarchical kind.

We mentioned coherence at the beginning of this post: in order to make the discussion precise, we need to define what we mean.

Mathematical Formulation of the Coherency Problem

To reason clearly, it helps to express your concepts in mathematical form. The key idea is simple: **all series in the hierarchy are linear combinations of the most granular ones**.

Let *yt* denote the vector containing *all* series in the hierarchy at time *t* - both aggregated and bottom-level - and let *bt* denote the vector of bottom-level series only. The relationship between them is fixed and known in advance, and it is fully described by the **summing matrix** *S (a.k.a. or aggregation matrix*).

This matrix encodes how bottom-level series add up to form higher-level aggregates. With this notation, the entire hierarchy can be written as:

$$y_t = S\ b_t$$

This equation expresses the **coherency constraint**: if the hierarchy is coherent, all series, at all levels, must satisfy this linear relationship.

Base forecasts and incoherence

In practice, forecasts are typically generated **independently** for each series in the hierarchy - so they almost never satisfy the aggregation constraint above. In other words, base forecasts are generally **incoherent**: the sum of lower-level forecasts does not match forecasts at higher levels.

The purpose of hierarchical forecasting is to **reconcile** these base forecasts into a new set of forecasts that **do** satisfy the hierarchy.

Linear reconciliation

A large and practically important class of reconciliation methods operates by applying a **linear transformation** to the base forecasts. The reconciled forecasts can always be written in the form:

$$\tilde{y} = S\ G\hat{y}$$

Here G is a *mapping matrix* that determines how information from all base forecasts is combined to produce revised bottom-level forecasts, which are then aggregated back up using S.

This simple linear framework is sufficient to describe all reconciliation methods discussed in this post.

Different hierarchical methods correspond to different choices of G:

- **Bottom-Up** uses a mapping that ignores all aggregated forecasts and keeps only bottom-level ones.
- **Top-Down** extracts information from the top level and redistributes it downward using fixed proportions.
- **Optimal reconciliation methods** (such as MinT) choose G to minimize forecast uncertainty while enforcing coherence.

This formulation provides a unifying view of hierarchical forecasting: no matter which method we choose, reconciliation means transforming incoherent base forecasts into coherent ones by respecting the fixed aggregation structure of the hierarchy.

Baseline

For illustration, we use a simplified subset of the M5 dataset: sales of a single product across all Walmart stores over the selected time period. This gives us a clean, interpretable hierarchy without losing the real-world complexity that makes hierarchical forecasting interesting.

We rely on the `hierarchicalforecast` library from Nixtla, which provides a practical implementation of both heuristic and optimal reconciliation methods. It requires some upfront preparation (*constructing base forecasts and a summing matrix*), but the resulting workflow is transparent and easy to experiment with.

First thing we do is create the baseline forecasts:

- **Define the hierarchy explicitly:** We start by specifying the hierarchical structure of the data - identifying the bottom-level series and how they aggregate into higher levels. This step is purely structural and independent of any forecasting model. What does the hierarchy look like in terms of nodes / values?

  ```
  tags['top_level'] : ['Total']
  ```

tags['top_level/middle_level'] : ['Total/CA' 'Total/TX' 'Total/WI']
tags['top_level/middle_level/bottom_level'] : ['Total/CA/CA_1' 'Total/CA/CA_2'
'Total/CA/CA_3' 'Total/CA/CA_4'
'Total/TX/TX_1' 'Total/TX/TX_2' 'Total/TX/TX_3' 'Total/WI/WI_1'
'Total/WI/WI_2' 'Total/WI/WI_3']

- **Prepare a consistent long-format dataset:** All series are reshaped into a common time-indexed format, with explicit identifiers for each level of the hierarchy. This ensures that forecasting and reconciliation operate on a unified representation of the data.
- **Split data into training and evaluation windows:** A clear cutoff is chosen to separate historical data used for model fitting from the forecast horizon. This mirrors a real forecasting workflow and allows fair comparison across reconciliation methods.
- **Generate independent base forecasts for every series:** Each node in the hierarchy (*both aggregated and bottom-level*) is forecast independently using the same univariate modeling approach - we pick AutoARIMA (see Episode 3 for a refresher). *At this stage, forecasts are intentionally incoherent.*
- **Treat base forecasts as inputs:** These initial forecasts serve as raw material for reconciliation. Their role is to capture local patterns at each level, while coherence and cross-series information are imposed later.

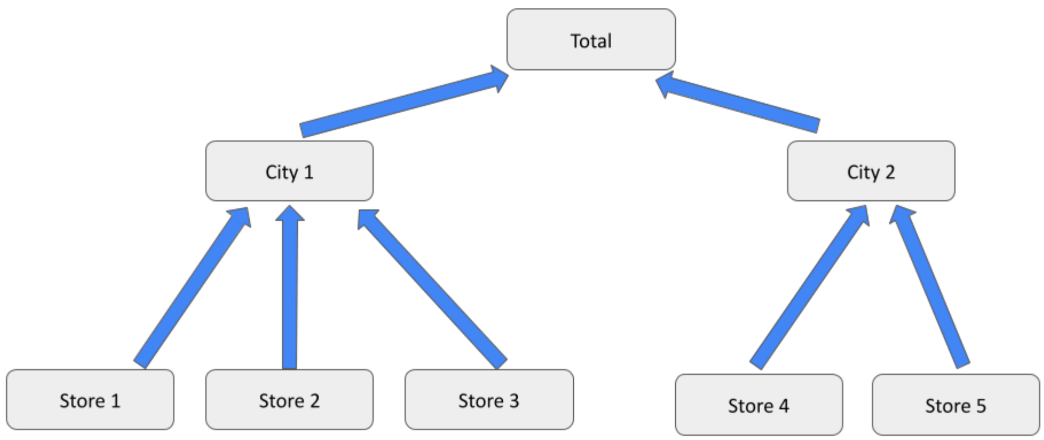
For reference, we evaluate the errors of those baseline forecast - before any reconciliation:

Overall rmse : 2.15
top_level rmse: 5.06
top_level/middle_level rmse: 2.45
top_level/middle_level/bottom_level rmse: 1.45

Aggregation methods: heuristics

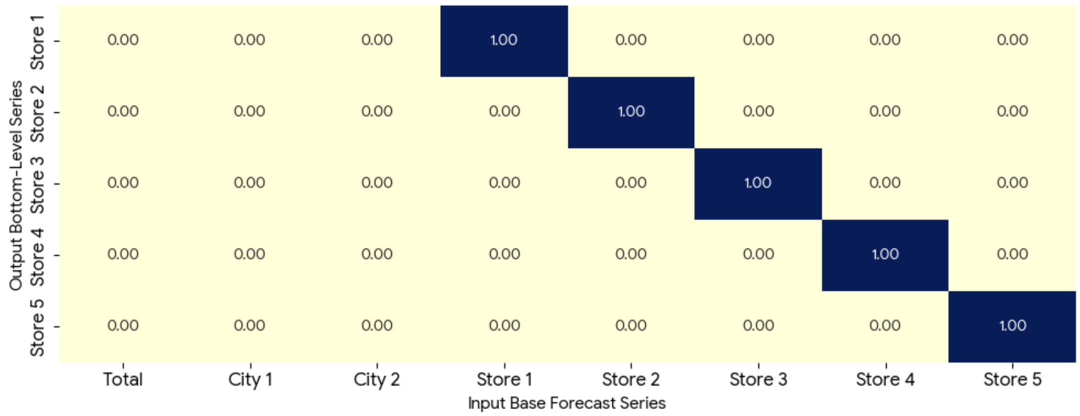
BottomUp

The Bottom-Up (BU) method is the most intuitive and widely used strategy in retail and supply chain forecasting. It involves forecasting only the lowest level of the hierarchy and aggregating these forecasts to produce estimates for all higher levels.



Mechanism:

- Train forecasting models for the bottom-level series.
- Generate base forecasts .
- Calculate coherent forecasts for the entire hierarchy using the summing matrix:



Advantages:

- **Granularity preservation:** It captures the rich dynamics of the lowest level. Idiosyncratic events (e.g. local store promotion or a regional holiday) are explicitly modeled and "flow up" to the aggregate forecast.
- **No information loss from aggregation:** Aggregation acts as a smoothing low-pass filter. By forecasting at the bottom, the model retains high-frequency signal that might be "averaged out" in the aggregate series.
- **Simplicity:** The method is mechanically straightforward and mathematically guaranteed to be coherent. It requires no complex matrix inversion or optimization.

Disadvantages:

- **Signal-to-Noise ratio:** Bottom-level data is often characterized by high volatility and intermittency (*lots of contiguous zeros*). Summing these noisy forecasts can lead to aggregate forecasts that are less accurate than if the aggregate had been forecast directly - *as errors may compound rather than cancel out.*
- **Missing macro trends:** It ignores broader trends, seasonality, or cyclicality that are only detectable at higher levels of aggregation; e.g. a subtle nationwide economic downturn might be invisible in

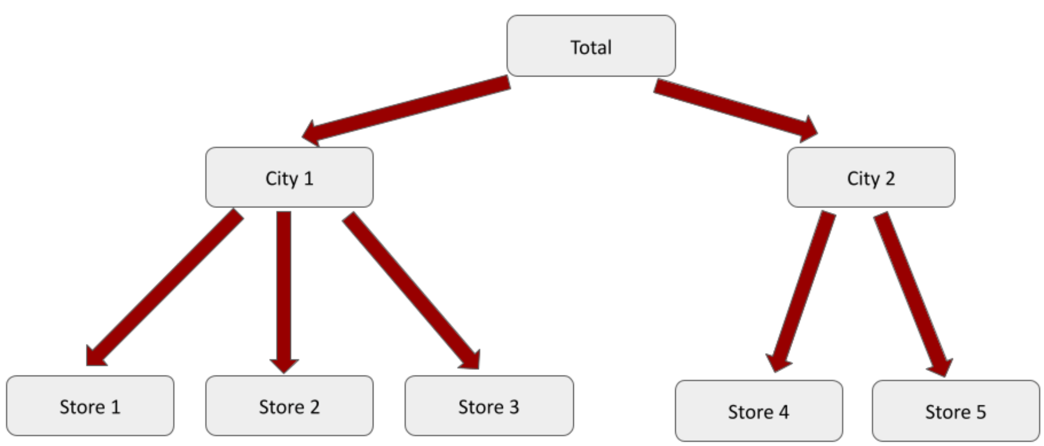
the noise of individual store sales but obvious in the national aggregate.

We now evaluate how this reconciliation strategy affects forecast accuracy across different levels of the hierarchy.

level	baseline_rmse	reconciled_rmse	delta_rmse	improved
top_level	5.06	5.14	0.08	False
top_level/middle_level	2.45	2.52	0.07	False
top_level/middle_level/bottom_level	1.45	1.45	0.00	False

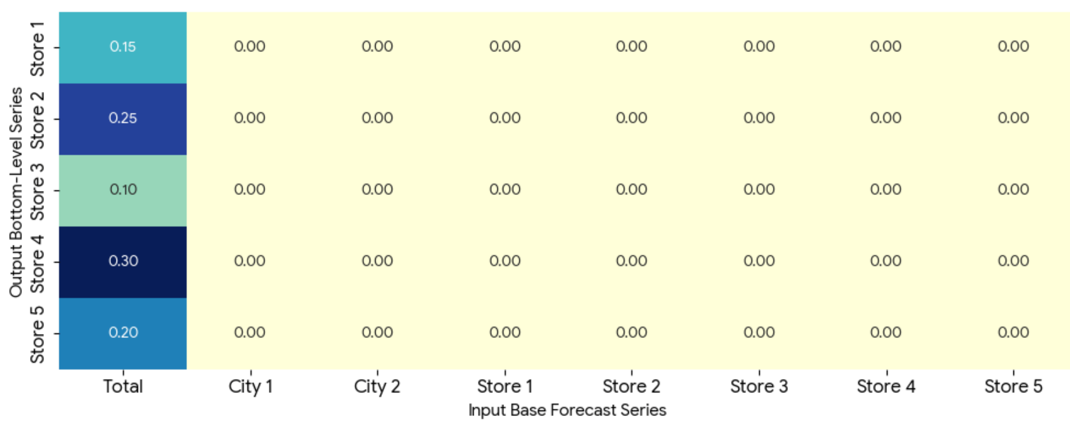
TopDown

The Top-Down (TD) method takes the opposite approach to Bottom-Up. Instead of forecasting the most granular series, it generates a forecast only for the top level of the hierarchy (*e.g. total sales*) and then disaggregates this forecast down to lower levels using fixed proportions derived from historical data.



Mechanism:

- Train a forecasting model for the top-level (aggregate) series.
- Generate a base forecast for the total.
- Estimate disaggregation proportions for each bottom-level series, typically based on historical averages or historical shares of the total.
- Distribute the top-level forecast to the bottom level using these proportions.
- Obtain forecasts for intermediate levels by aggregating the reconciled bottom-level forecasts.



Advantages:

- **Stability:** Aggregate series are typically much smoother than bottom-level data due to noise cancellation. This makes the top-level forecast easier to model and often more accurate in isolation.
- **Computational efficiency:** Only a single forecasting model needs to be trained and maintained, making Top-Down attractive in large-scale or resource-constrained settings.
- **Robustness to sparsity:** When bottom-level series are intermittent or extremely noisy, Top-Down can outperform Bottom-Up by avoiding direct modeling of unreliable signals.

Disadvantages:

- **Loss of heterogeneity:** Top-Down assumes that all sub-series evolve proportionally. If the total grows or shrinks, all components inherit this change, even when individual dynamics differ.
- **Static structure assumption:** Disaggregation proportions are often estimated from historical averages and assumed to be stable over time, *which may fail in the presence of structural change*.
- **Limited local sensitivity:** Local effects (e.g. regional promotions or store-specific trends) are not modeled directly and can be missed at lower levels.

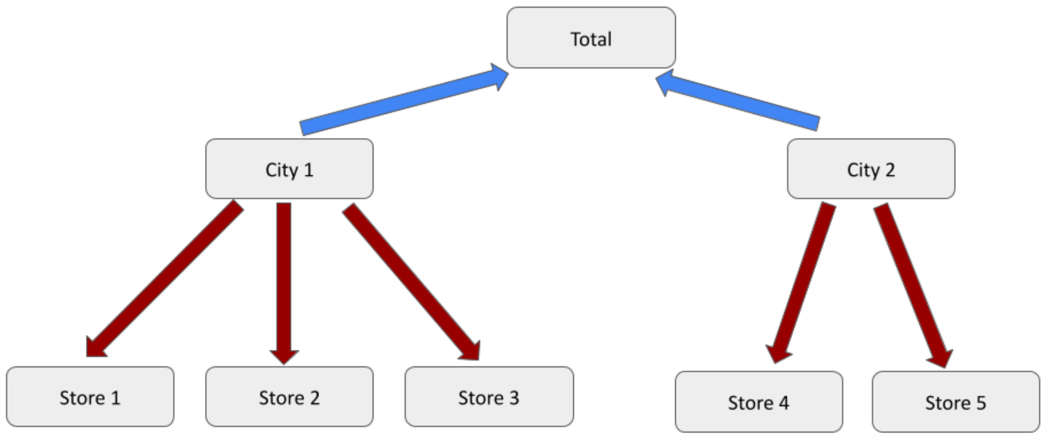
How does TopDown perform as a reconciliation method?

level	baseline_rmse	reconciled_rmse	delta_rmse	improved
top_level	5.06	5.06	0.00	False
top_level/middle_level	2.45	2.43	-0.02	True
top_level/middle_level/bottom_level	1.45	1.44	-0.01	True

Better than BottomUp - top level is unchanged (as expected ;-) but both middle and bottom ones improve. This is notable, since the default setting (= historical proportions) is a fairly simple heuristic.

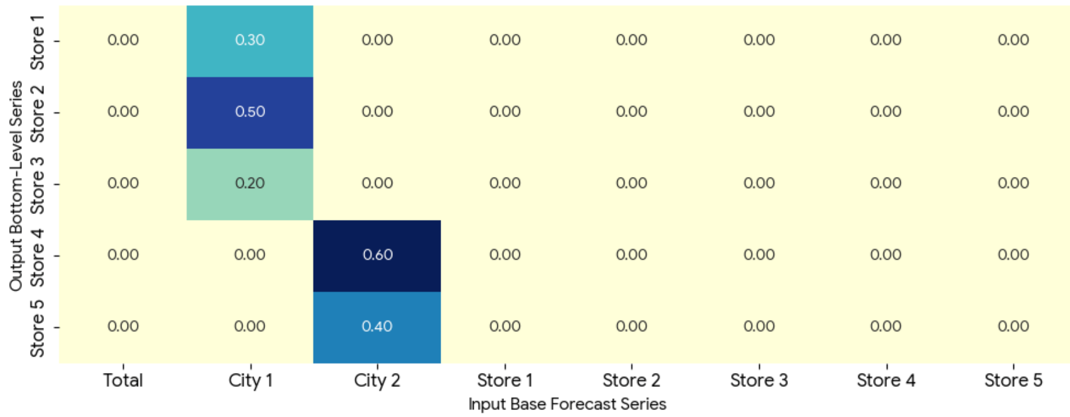
MiddleOut

The Middle-Out (MO) method is a hybrid strategy that aims to balance the stability of aggregated forecasts with the detail of bottom-level modeling. Instead of anchoring forecasts at the top or the bottom of the hierarchy, a middle level is selected - often the level at which decisions are made or where the signal-to-noise ratio is most favorable.



Mechanism:

- Select an intermediate level of the hierarchy (e.g., “Region” in a Country–Region–Store structure).
- Train forecasting models for all series at the chosen middle level.
- Generate base forecasts for the middle-level series.
- Obtain forecasts for higher levels by aggregating the middle-level forecasts (Bottom-Up logic).
- Obtain forecasts for lower levels by disaggregating the middle-level forecasts using historical proportions (Top-Down logic)



Advantages:

- **Balanced signal quality:** Middle-level series are often less noisy than bottom-level data while retaining more structure than fully aggregated series.
- **Operational alignment:** The chosen level frequently corresponds to how organizations plan and make decisions, improving interpretability and usability.
- **Reduced modeling burden:** Compared to Bottom-Up, fewer models need to be trained, while still capturing more heterogeneity than pure Top-Down.

Disadvantages:

- **Subjective level selection:** Choosing the "right" middle level relies heavily on domain knowledge and may not be obvious or stable over time.
- **Inherited limitations:** The method still suffers from Top-Down weaknesses when disaggregating downward (loss of local detail) and Bottom-Up weaknesses when aggregating upward (error accumulation).
- **Static proportions:** As with Top-Down, disaggregation often assumes stable historical relationships, which can break under

structural change.

What is the impact of MO on our base forecast reconciliation?

level	baseline_rmse	reconciled_rmse	delta_rmse	improved
top_level	5.06	5.14	0.08	False
top_level/middle_level	2.45	2.52	0.07	False
top_level/middle_level/bottom_level	1.45	1.45	0.00	False

For our particular use case, MO makes things uniformly worse - on all levels of the hierarchy.

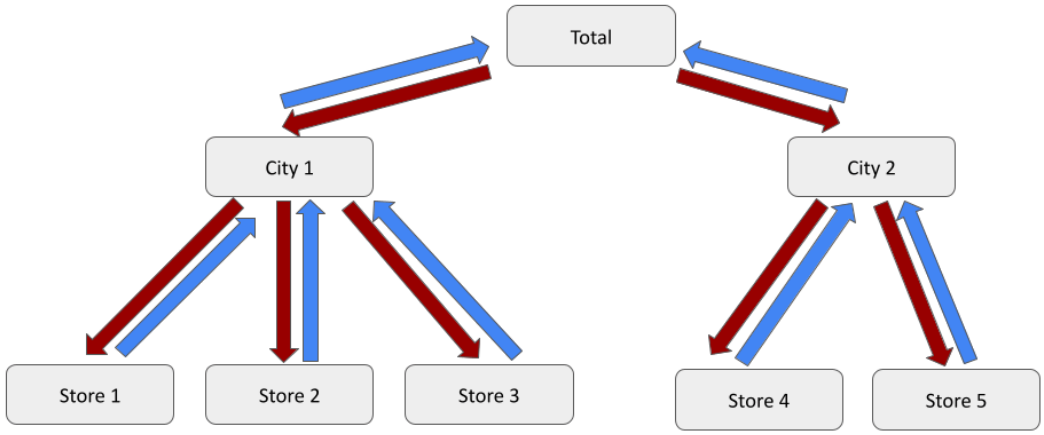
The heuristic methods discussed so far restrict how information flows through the hierarchy: Bottom-Up ignores aggregate-level signals, Top-Down ignores local dynamics, and Middle-Out fixes a single compromise point. While these approaches are simple and often effective, they all discard potentially useful information.

Optimal reconciliation methods remove this restriction: instead of choosing *which* level to trust, they combine information from **all levels simultaneously**, using the historical error structure of the forecasts to determine how much weight each series should receive. This way strong aggregate trends can help stabilize noisy bottom-level forecasts, while local signals can correct overly smooth totals.

Optimal Reconciliation

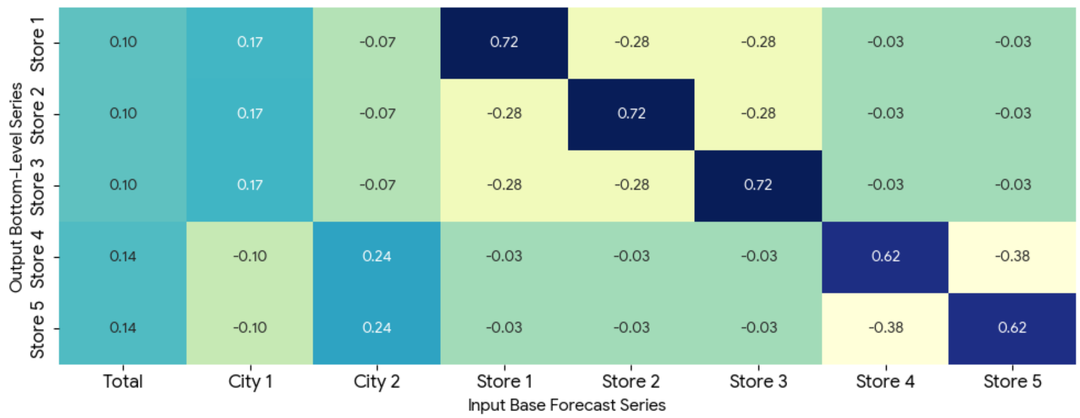
MinTrace

The Minimum Trace (MinT) method is an optimal reconciliation approach that produces coherent forecasts by minimizing the overall uncertainty of the reconciled hierarchy. MinT uses the covariance structure of forecast errors to determine how information should be shared across levels. *In practice, variants such as OLS, WLS, and shrinkage estimators are used to balance optimality and robustness.*



Mechanism:

- Generate independent base forecasts for **all** series in the hierarchy.
- Estimate the covariance matrix of the base forecast errors using historical residuals.
- Construct a reconciliation mapping that minimizes the total variance of the reconciled forecasts, subject to the coherence constraints imposed by the hierarchy.
- Apply this mapping to the base forecasts to obtain coherent forecasts at all levels.



Advantages:

- **Statistical optimality:** Under standard assumptions (notably unbiased base forecasts), MinT produces coherent forecasts with the minimum total forecast variance.
- **Information sharing across levels:** Signals from stable aggregate series and detailed bottom-level series are combined automatically.
- **Unified framework:** Bottom-Up and Top-Down can be recovered as special cases, making MinT a generalization.

How does it impact the base forecasts for our use case?

level	baseline_rmse	reconciled_rmse	delta_rmse	improved
top_level	5.06	5.07	0.01	False
top_level/middle_level	2.45	2.45	0.00	False
top_level/middle_level/bottom_level	1.45	1.44	-0.01	True

Minor improvement at the lowest level, but overall inferior to TopDown.

Closing time

Hierarchical forecasting has evolved from a bookkeeping exercise into a central modeling problem in modern forecasting systems. What begins as a simple question *"which level should we forecast?"* quickly turns into a set of trade-offs involving noise, bias, and how information should flow across the hierarchy.

Simple heuristic approaches such as Bottom-Up, Top-Down, and Middle-Out remain valuable. They are transparent, fast, and often sufficient in practice - especially when data quality or operational constraints limit more sophisticated methods. Optimal reconciliation techniques like MinT provide a principled alternative: they combine information from all levels at once - but at the cost of stronger assumptions and higher computational overhead.

In practice, there is no universally best approach: data sparsity, hierarchy depth, forecasting horizon, and business objectives all matter. Increasingly, large-scale global models blur these distinctions by learning shared structure directly across series -
a topic we will return to in later episodes.

Key Takeaways

1. **No Free Lunch:** No single reconciliation method dominates in all settings. Bottom-Up is simple and cheap but sensitive to noise; MinT is theoretically optimal but computationally heavier and sensitive to covariance estimation.
2. **Heuristics are still useful:** Bottom-Up, Top-Down, and Middle-Out remain strong baselines. Their transparency and low operational cost often make them the right choice in real-world systems.
3. **Optimal reconciliation trades assumptions for efficiency:** MinT improves information sharing across the hierarchy by exploiting error correlations, *but its performance depends on the quality of base forecasts and error estimates*.
4. **Data quality drives method choice:** Sparse or intermittent bottom-level data favors Top-Down or Middle-Out. Rich, stable data favors Bottom-Up or MinT-style approaches.

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